PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
FORM FOUR KIBAHA PRIVATE SECONDARY SCHOOL ASSOCIATION (KIPSSA)

033/2 BIOLOGY 2A (ACTUAL PRACTICAL)
Time: 2:30 Hours Year: 2025

Instructions.

1. This paper consists of **two (2)** questions, answer all questions.
2. This paper carries **fifty (50)** Marks.
3. All writings must be written in a **blue** or **black** ink except drawings must be in pencil.
4. All necessary works and answers for each question must be shown clearly.
5. Write your **Examination Number** on every page of your answer booklet(s) provided.

QUESTIONS.

1. You are provided with specimen **Y** which has been exposed to direct sunlight for 5 hours while attached to the mother plant. you are provided with methylated spirit and iodine solution. Use the to perform the experiment in procedures (i)-(iv),then answer the question that follow:

Procedure

(i) Put specimen **Y** in the beaker and boil it for 3minutes, then remove it.
(ii) Dip the boiled specimen **Y** into test tube containing methylated spirit and place the test tube in a beaker of boiling water for about 5 minutes. Observe the change
(iii) Remove specimen **Y** from hot methylated spirit and rinse it with warm water. Note the texture of the specimen when removed from hot methylated spirit and after rinsing with warm water.
(iv) Put rinsed specimen **Y** on a white tile and add 3 drops of iodine solution on the surface of the specimen. Observe the changes

Questions

(a) What was the aim the experiment?

(b) Why specimen **Y** was boiled in water for about 4minutes?

(c) Identify the observation made when specimen **Y** boiled in methylated spirit.

(d) Briefly explain one importance for each of the following :
   (i) Putting specimen **Y** in hot methylated spirit.
   (ii) Placing a test tube containing specimen **Y** and methylated spirit in a beaker of boiling water.

(e) Identify the texture of specimen **Y** after:
   (i) Removing the specimen from hot methylated spirit.
   (ii) Rinsing the specimen with warm water.

(f) Give your observation when iopdine solution was added on the surface of specimen **Y**

(g) What is your conclusion about the results of the experiment in procedures (i)-(iv)?

(h) Draw a diagram of specimen **Y** and label four external parts.

2. Study the given specimens **K1, K2, K3, K4**, and **K5** and then answer the questions that follow.

a) i).Give the common names of the given specimen
   ii) State any four adaptations of specimen K2 to its environment.

b) i) Draw a well labeled diagram of specimen K1.
   ii)Classify specimen **K1,K2,K3,K4** and **K5** to their class level.
   iii) Sate any two economic importance for each specimen **K1** and **K5**

c) i) State any 3 reasons for placing specimen **K3** and **K4** to the same kingdom
   ii) What are 4 observable differences between specimen **K3** and **K4**.
   iii) Define the type of germination showed by specimen **K3**.
   iv) Draw a well labeled diagram of specimen **K4**.
   v) State economic importance of specimen **K4** to human being.